



MASTER'S DEGREE IN DATA SCIENCE

1. EXECUTIVE PG PROGRAMME FROM IIITB

PRE-PROGRAM PREPARATORY CONTENT

Module

- DATA ANALYSIS IN EXCEL

Description

Taught by one of the most renowned data scientists in the country (S.Anand, CEO, Gramener), this module takes you from a beginner level Excel user to an almost professional user.

- ANALYTICS PROBLEM SOLVING

Description

This module covers concepts of the CRISP-DM framework for business problem-solving.

COURSE 1 - DATA TOOLKIT

2 ASSIGNMENTS

Module

- INTRODUCTION TO PYTHON - I

Description

Build a foundation for the most in-demand programming language of the 21st century.

- INTRODUCTION TO PYTHON - II

Description

Learn to apply some of the commonly used paradigms of functional programming in Python.

- PROGRAMMING IN PYTHON

Description

Learn how to approach and solve logical problems using programming.

- DATA ANALYSIS USING SQL

Description

Data in companies is definitely not stored in excel sheets! Learn the fundamentals of database and extract information from RDBMS using the structured query language.



- PYTHON FOR DATA SCIENCE

Description

Learn how to manipulate datasets in Python using Pandas which is the most powerful library for data preparation and analysis.

- VISUALISATION IN PYTHON

Description

Humans are visual learners and hence no task related to data is complete without visualisation. Learn to plot and interpret various graphs in Python and observe how they make data analysis and drawing insights easier.

- EXPLORATORY DATA ANALYSIS

Description

Learn how to find and analyse the patterns in the data to draw actionable insights.

- IMDB MOVIE ASSIGNMENT

Description

Reinforce the concepts learnt in data science through this rigorous assignment involving the past.

- MATHS FOR DATA SCIENCE

Description

Build the mathematical foundation required for understanding the Machine Learning Algorithms.

- INFERENCE STATISTICS

Description

Build a strong statistical foundation and learn how to 'infer' insights from a huge population using a small sample.

- HYPOTHESIS TESTING

Description

Understand how to formulate and validate hypotheses for a population to solve real-life business problems.

- ADVANCED SQL

Description

Apply advanced SQL concepts like windowing and procedures to derive insights from data and answer pertinent business questions.

- CREDIT EDA CASE STUDY

Description

Solve a real industry problem through the concepts learnt in exploratory data analysis.



Module

- INTRODUCTION TO MACHINE LEARNING AND LINEAR REGRESSION

Description

Venture into the machine learning community by learning how one variable can be predicted using several other variables through a housing dataset where you will predict the prices of houses based on various factors.

- LINEAR REGRESSION ASSIGNMENT - BIKE SHARING SYSTEMS

Description

Build a model to understand the factors on which the demand for bike sharing systems vary on and help a company optimise its revenue.

- LOGISTIC REGRESSION

Description

Learn your first binary classification technique by determining which customers of a telecom operator are likely to churn versus who are not to help the business retain customers.

- UNSUPERVISED LEARNING: CLUSTERING

Description

Learn how to group elements into different clusters when you don't have any pre-defined labels to segregate them through K-means clustering, hierarchical clustering, and more.

- BUSINESS PROBLEM SOLVING

Description

Learn how to approach open ended real world problems using data as a lever to draw actionable insights.

- CLUSTERING ASSIGNMENT (OPTIONAL)

Description

Apply the machine learning concepts learnt to help an international NGO cluster countries to determine their overall development and plan for lagging countries.

- CASE STUDY: LEAD SCORING

Description

Help the Sales team of your company identify which leads are worth pursuing through this classification case study.



SPECIALISATION 1: DATA SCIENCE GENERALIST

ADVANCED MACHINE LEARNING AND STORYTELLING

3 ASSIGNMENTS

Module

- TREE MODELS + BOOSTING (OPTIONAL)

Description

Learn how the human decision making process can be replicated using a decision tree and other powerful ensemble algorithms.

- MODEL SELECTION & GENERAL ML TECHNIQUES

Description

Learn the pros and cons of simple and complex models and the different methods for quantifying model complexity, along with general machine learning techniques like feature engineering, model evaluation, and many more.

- ML LAB I: CLASSIFICATION

- PRINCIPAL COMPONENT ANALYSIS

Description

Understand important concepts related to dimensionality reduction, the basic idea and the learning algorithm of PCA, and its practical applications on supervised and unsupervised problems.

- ADVANCED REGRESSION - I

Description

In this module, take a more advanced look at regression models and learn the concepts related to regularisation.

- ADVANCED REGRESSION - II & ML LAB II: REGRESSION

- TEXT ANALYTICS & PROCESSING + TEXT-BASED PREDICTIVE MODELLING

Description

An introduction to the world of NLP and basic text processing skills. Learn how to build a classification engine that works on (unstructured) textual data.

- BASIC VISUALISATION USING TABLEAU

Description

Learn advanced visualisation techniques using the most in-demand visualisation tool in the industry.



- DATA STORYTELLING

Description

Learn how to effectively strategise, communicate, and fine grain your data analysis projects and understand how to optimally present your findings to technical and non-technical stakeholders and upgrade your storytelling skills.

- BUSINESS CASE STUDY

ADVANCED PROGRAMMING AND DATABASES

2 ASSIGNMENTS

Module

- DATA MODELLING

Description

In this module, you will learn and use data modelling on a dataset to solve a business problem.

- SQL WEEKLONG LAB
- ADVANCED SQL - WEEK II

Description

Apply advanced SQL concepts like windowing and procedures to derive insights from data and answer pertinent business questions

- ALGORITHM ANALYSIS + RECURSION

Description

Learn how to assess the efficiency your code using algorithm analysis techniques and learn to write recursive algorithms

- SEARCHING AND SORTING (DIVIDE AND CONQUER INCLUDED)

Description

Learn most fundamental searching and sorting algorithms and design techniques

- DATA STRUCTURES - SETS, DICTIONARIES, STACKS, QUEUES

Description

Learn user defined data structures -Stack, Queue, Trees in Python that help in advanced data manipulation

- PYTHON - OOPS

Description

Learn OOP concepts such as Class, Object, Method, Inheritance, Polymorphism, Data Abstraction and Encapsulation.



- PYTHON WEEKLONG LAB

CAPSTONE PROJECT WITH VIDEO PRESENTATION

Module

- CAPSTONE PROJECT

Description

Solve an end-to-end real-life industry problem from a wide variety of domains. Make a video presentation of your working demo to showcase in your portfolio.

SPECIALISATION 2: NATURAL LANGUAGE PROCESSING

ADVANCED MACHINE LEARNING

2 ASSIGNMENTS

Module

- TREE MODELS

Description

Learn how the human decision making process can be replicated using a decision tree and other powerful ensemble algorithms.

- MODEL SELECTION & GENERAL ML TECHNIQUES

Description

Learn the pros and cons of simple and complex models and the different methods for quantifying model complexity, alongwith general machine learning techniques like feature engineering, model evaluation, and many more.

- PRINCIPAL COMPONENT ANALYSIS

Description

Understand important concepts related to dimensionality reduction, the basic idea and the learning algorithm of PCA, and its practical applications on supervised and unsupervised problems.

- ADVANCED REGRESSION

Description

In this module, take a more advanced look at regression models and learn the concepts related to regularisation.

- ADVANCED REGRESSION ASSIGNMENT

Description

Build a regularised regression model to understand the most important variables to predict the house prices in Australia.



- BAGGING AND BOOSTING

Description

Learn about ensemble modelling through bagging and boosting and understand how weak algorithms can be transformed into stronger ones.

- TIME SERIES ANALYSIS

Description

In this module, you will learn how to analyse and forecast a series that varies with time.

- TELECOM CHURN CASE STUDY

Description

Solve the most crucial business problem for a leading telecom operator in India and southeast Asia - predicting customer churn.

NATURAL LANGUAGE PROCESSING

2 ASSIGNMENTS

Module

- TREE MODELS

Description

Do you get annoyed by the constant spams in yor mail box? Wouldn't it be nice if we had a program to check your spellings? In this module learn how to build a spell checker & spam detector using techniques like phonetic hashing, bag-of-words, TF-IDF, etc.

- FEATURE EXTRACTION & MODELLING

Description

This module will help you in understanding how to extract meaningful features from the processed text data. Using these features you will be able to extract entities, classify POS tags, generating similarity score between two question strings.

- ASSIGNMENT - NATURAL LANGUAGE PROCESSING

Description

To perform a sentiment analysis on product reviews from Amazon using NLP & Machine learning. This assignment will be focused to give you a business understanding on how to do product optimisation using NLP.

- INTRO TO DL

Description

Learn the most sophisticated and cutting-edge technique in machine learning - Artificial Neural Networks & how to apply Deep learning for NLP.



- NLP INDUSTRIAL APPLICATIONS

Description

Learn how to use NLP with Neural networks for different industrial applications like text classification, Question pair similarity, Text generation, Topic modelling.

- CHATBOT CASE STUDY

Description

Imagine if you could make a restaurant booking without opening Zomato. Build your own restaurant search chatbot with the help of RASA - an open source framework and deploy it on Slack.

CAPSTONE

Module

- DEPLOYMENT

Description

Learn how to productionise your model and deploy it on the server.

- CAPSTONE

Description

Choose from a range of real-world industry woven projects on advanced topics like Recommendation Systems, Fraud Detection, Emotion Detection from faces, Social Media Listening, Speech Recognition among many others.

SPECIALISATION 3: DEEP LEARNING

ADVANCED MACHINE LEARNING

2 ASSIGNMENTS

Module

- TREE MODELS

Description

Learn how the human decision making process can be replicated using a decision tree and other powerful ensemble algorithms.

- MODEL SELECTION & GENERAL ML TECHNIQUES

Description

Learn the pros and cons of simple and complex models and the different methods for quantifying model complexity, along with general machine learning techniques like feature engineering, model evaluation, and many more.



- PRINCIPAL COMPONENT ANALYSIS

Description

Understand important concepts related to dimensionality reduction, the basic idea and the learning algorithm of PCA, and its practical applications on supervised and unsupervised problems.

- ADVANCED REGRESSION

Description

In this module, take a more advanced look at regression models and learn the concepts related to regularisation.

- ADVANCED REGRESSION ASSIGNMENT

Description

Build a regularised regression model to understand the most important variables to predict the house prices in Australia.

- BAGGING AND BOOSTING

2 ASSIGNMENTS

Description

Learn about ensemble modelling through bagging and boosting and understand how weak algorithms can be transformed into stronger ones.

- TIME SERIES ANALYSIS

Description

In this module, you will learn how to analyse and forecast a series that varies with time.

- TELECOM CHURN CASE STUDY

Description

Solve the most crucial business problem for a leading telecom operator in India and southeast Asia - predicting customer churn.

DEEP LEARNING AND NEURAL NETWORKS

2 ASSIGNMENTS

Module

- INTRODUCTION TO NEURAL NETWORKS

Description

Learn the most sophisticated and cutting-edge technique in machine learning - Artificial Neural Networks or ANNs.



- NEURAL NETWORKS ASSIGNMENT

Description

Build a neural network from scratch in Numpy to identify handwritten digits.

- CONVOLUTIONAL NEURAL NETWORKS - INTRODUCTION AND INDUSTRY APPLICATIONS

Description

Learn the basics of CNN and OpenCV and apply it to Computer Vision tasks like detecting anomalies in chest X-Ray scans, vehicle detection to count & categorise them to help the government ascertain the width and strength of the road.

- RECURRENT NEURAL NETWORKS

Description

Ever wondered what goes behind machine translation, sentiment analysis, speech recognition? Learn how RNN helps in these areas having sequential data like text, speech, videos, and a lot more.

- GESTURE RECOGNITION

Description

Make a Smart TV system which can control the TV with user's hand gestures as the remote control.

CAPSTONE

Module

- DEPLOYMENT

Description

Learn how to productionise your model and deploy it on the server.

- CAPSTONE

Description

Choose from a range of real-world industry woven projects on advanced topics like Recommendation Systems, Fraud Detection, Emotion Detection from faces, Social Media Listening, Speech Recognition among many others.





SPECIALISATION 4: BUSINESS ANALYTICS

ADVANCED MACHINE LEARNING

2 ASSIGNMENTS

Module

- TREE MODELS

Description

Learn how the human decision making process can be replicated using a decision tree and other powerful ensemble algorithms.

- TIME SERIES FORECASTING

Description

In this module, you will learn how to analyse and forecast a series that varies with time.

- RETAIL-GIANT SALES FORECASTING ASSIGNMENT

Description

Apply the concepts learnt in time series to solve a forecasting problem for a retail giant.

- MODEL SELECTION & GENERAL ML TECHNIQUES

Description

Learn the pros and cons of simple and complex models and the different methods for quantifying model complexity, along with general machine learning techniques like feature engineering, model evaluation, and many more.

- SQL BEST PRACTICES

Description

Learn how to write optimised SQL query that require less memory and execute in lesser amount of time.

- ADVANCED EXCEL

Description

Learn the advanced concepts in Excel and start to perform data analysis like a pro!

- TELECOM CHURN CASE STUDY

Description

Solve the most crucial business problem for a leading telecom operator in India and southeast Asia - predicting customer churn.



BUSINESS REQUIREMENTS

2 ASSIGNMENTS

Module

- STRUCTURED PROBLEM SOLVING USING FRAMEWORKS

Description

Learn how to attack a business problem using various structured frameworks like 5W, 5WHYs, and SPIN.

- STRUCTURED PROBLEM SOLVING ASSIGNMENT

Description

Apply your learnings from the course to solve a real-life business problem.

- OPERATIONS RESEARCH

Description

Learn about the world of operations research through linear and integer optimisations.

- DATA STORYTELLING

Description

Learn how to effectively strategise, communicate, and fine grain your data analysis projects and understand how to optimally present your findings to technical and non-technical stakeholders and upgrade your storytelling skills.

- OPERATIONS RESEARCH CASE STUDY

Description

Understand how a project in the industry is taken up and solved through a comprehensive business case study.

CAPSTONE PROJECT

5 ASSIGNMENTS

Module

- CAPSTONE PROJECT

Description

Solve an end-to-end real-life industry problem from a wide variety of domains. Available capstone project choices -

- (i) Stock Analysis & Portfolio Management
- (ii) E-Commerce & Marketing
- (iii) Healthcare
- (iv) Supply Chain Optimisation
- (v) Credit Card Fraud Detection



SPECIALISATION 5: BUSINESS INTELLIGENCE / DATA ANALYTICS

SQL AND NOSQL DATABASES

2 ASSIGNMENTS

Module

- DATA MODELLING

Description

In this module, you will learn and use data modelling on a dataset to solve a business problem.

- SQL BEST PRACTICES

Description

Learn how to write optimised SQL query that require less memory and execute in lesser amount of time.

- SQL ASSIGNMENT: IMDB MOVIES

Description

In this assignment, you will work on a movies dataset using SQL to extract exciting insights.

- ADVANCED EXCEL

Description

Learn the advanced concepts in Excel and start to perform data analysis like a pro!

- NOSQL DATABASES AND BEST PRACTICES

Description

Take your knowledge of query languages a step further by learning about MongoDB - a NoSQL database which is becoming more and more popular in the industry.

- INTRODUCTION TO BIG DATA AND CLOUD

Description

Understand the basics of big data and cloud and learn to work with an EMR cluster on a cloud based service.

- HIVE AND QUERYING

Description

In this module, you learn about the architecture and features of the Hive Query Language.

- HIVE CASE STUDY

Description

Understand how a project in the industry is taken up and solved through a comprehensive business case study.



STORYTELLING WITH ADVANCED VISUALISATIONS

2 ASSIGNMENTS

Module

- VISUALISATION USING TABLEAU

Description

Learn advanced visualisation techniques using the most in-demand visualisation tool in the industry.

- SPORTS ANALYTICS - IPL VISUALISATION ASSIGNMENT

Description

Apply the new found Excel and Tableau skills to solve an exciting business assignment.

- VISUALISATION USING POWERBI

Description

Take your visualisation game a step forward by understanding how to operate PowerBI.

- VISUALISATION USING PLOTLY

Description

Get a brief introduction to another popular open-sourced visualisation library in Python and learn to code and create powerful, pretty, and interactive visualisations.

- DATA STORYTELLING

Description

Learn how to effectively strategise, communicate, and fine grain your data analysis projects and understand how to optimally present your findings to technical and non-technical stakeholders and upgrade your storytelling skills.

- PLOTLY CASE STUDY

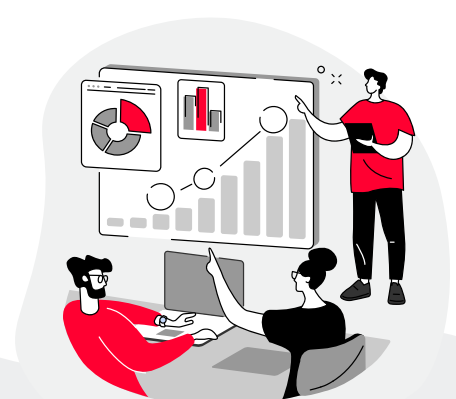
Description

Understand how a project in the industry is taken up and solved through a comprehensive business case study.

CAPSTONE PROJECT

5 ASSIGNMENTS

Module





- CAPSTONE PROJECT

Description

Solve an end-to-end real-life industry problem from a wide variety of domains. Available capstone project choices -

- (i) Web & Social Media Analytics
- (ii) Finance and Risk Analytics
- (iii) Marketing and Retail Analytics
- (iv) Supply Chain Analytics
- (v) Fraud Analytics

SPECIALISATION 6: DATA ENGINEERING

DATA ENGINEERING I 4 ASSIGNMENTS (1 MANDATORY, 3 OPTIONAL)

Module

- INTRODUCTION TO BIG DATA(OPTIONAL)

Description

This module you will learn what big data is, its various characteristics, and its determining factors. You will also get an idea of the various sources of big data and the wide range of big data applications in different industries such as retail, healthcare, and finance.

- INTRODUCTION TO CLOUD AND AWS SETUP

Description

Understand what is cloud and setup your AWS account which will be required during the program.

- INTRODUCTION TO HADOOP AND MAPREDUCE PROGRAMMING

Description

Understand the world of distributed data processing and storage with Hadoop. Learn to write MapReduce jobs in Python.

- MAPREDUCE PROGRAMMING ASSIGNMENT (OPTIONAL)

Description

Practise MapReduce Programming on a Big Dataset.

- DATA MANAGEMENT AND RELATIONAL DATABASE MODELLING

Description

Understand the concepts of Data Management and learn to model data from a Relational Database.



- NOSQL DATABASES AND APACHE HBASE NOSQL DATABASES AND MONGODB (OPTIONAL)

Description

Learn the concepts of NoSQL databases. Understand the working of Apache HBase.

- DATA WAREHOUSING (OPTIONAL)

Description

Understand the intricacies behind designing a data warehouse and a data lake for use case/s.

- DATA INGESTION WITH APACHE SQOOP AND APACHE FLUME

Description

Get familiar with the challenges involved in data ingestion. Use Sqoop and Flume to ingest structured and unstructured data into Hadoop.

- HIVE & QUERYING

Description

Manage and query a data warehouse with Apache Hive. Learn to write optimised HQL for large scale data analysis.

- HIVE ASSIGNMENT (OPTIONAL)

Description

Use HQL to analyse a Big Dataset

- AMAZON REDSHIFT

Description

Learn to deploy a Redshift cluster and use it for querying data.

- INTRODUCTION TO APACHE SPARK

Description

Get introduced to Apache Spark, a lightning fast big data processing engine.

- NYC PARKING ASSIGNMENT (OPTIONAL)

Description

Practise Apache Spark and its core libraries on the NYC Parking Ticket dataset.

- PROJECT: ETL DATA PIPELINE

Description

Make use of Sqoop, Redshift & Spark to design an ETL data pipeline.



DATA ENGINEERING - II 3 ASSIGNMENTS (1 MANDATORY, 2 OPTIONAL)

Module

- OPTIMISING SPARK FOR LARGE SCALE DATA PROCESSING

Description

Use PySpark to create large scale data processing applications.

- APACHE FLINK(OPTIONAL)

Description

Get Introduced to Apache Flink and learn query batch data. Use DataStream API to create a stream processing application.

- REAL-TIME DATA STREAMING WITH APACHE KAFKA

Description

Understand the producer-consumer architecture of Apache Kafka. Learn to set up a Kafka cluster for managing real-time data.

- REAL-TIME DATA PROCESSING USING SPARK STREAMING

Description

Learn about the real-time data processing architecture of Apache Spark. Build Spark Streaming applications to process data in real-time.

- STOCK DATA ANALYSIS ASSIGNMENT (OPTIONAL)

Description

This assignment revolves around building Spark structured streaming application to processing stock data in real-time.

- BUILDING AUTOMATED DATA PIPELINES WITH AIRFLOW

Description

Automate Data Pipelines with Airflow.

- ANALYTICS USING PYSPARK

Description

Use PySpark to do EDA and Predictive Analysis using Spark's ML library.

- CLASSIFICATION ASSIGNMENT (OPTIONAL)

Description

An assignment related to a classification based problem statement.



- PROJECT: REAL TIME DATA PROCESSING

Description

Build an end-to-end real-time data processing application using Spark Streaming and Kafka.

CAPSTONE PROJECT

4 ASSIGNMENTS

Module

- CAPSTONE PROJECT

Description

The capstone project will stitch all the components of data engineering together.

SEMESTER 1 AT IU, GERMANY

Module

ECTS

- | | |
|--|----|
| • Cyber Security and Data Protection | 5 |
| • Case Study: Model Engineering | 5 |
| • Software Engineering for Data Intensive Sciences | 5 |
| • Electives B | 10 |
| • Seminar: Current Topics in Data Science | 5 |

SEMESTER 2 AT IU, GERMANY

Module

ECTS

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|------------------------------|----|
| • Master Thesis & Colloquium | 30 |
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ELECTIVE B OPTIONS

Module

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| • Software Engineering for Data Intensive Sciences |
| • Management |
| • Sales, Pricing and Brand Management |
| • Consumer Behavior and Research |



- Corporate Finance
- Innovate and Change
- Cognitive Computing
- Applied Autonomous Driving
- Self Learning Systems
- Industrial Automation and Internet of Things

